

JEDDAH KING ABDULAZIZ INTL, Saudi Arabia

WMO: 410240

Lat: 21.680N

Lon: 39.157E

Elev: 15

StdP: 101.15

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.2	17.2	1.1	4.1	26.6	4.2	5.1	26.9	11.2	25.7	10.1	25.3	2.6	30	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	41.0	23.7	39.9	24.2	38.9	24.6	29.8	35.0	29.1	34.4	28.5	33.9	6.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.8	25.4	33.5	27.9	24.1	32.7	27.1	22.9	32.3	99.6	35.3	95.5	34.8	93.0	33.6	35.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.6	8.5	7.7	14.4	45.6	1.2	2.2	13.5	47.1	12.8	48.4	12.1	49.6	11.2	51.2
			9.3	31.3	1.1	1.6	8.5	32.5	7.8	33.5	7.2	34.4	6.4	35.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.1	23.9	24.6	25.9	28.6	31.0	32.0	33.4	33.6	32.2	30.6	27.9	25.5
	DBStd	3.83	2.09	2.23	2.34	2.40	2.18	1.78	1.33	1.29	1.32	1.35	1.69	1.80
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6981	431	408	493	557	651	659	726	731	666	639	538	482
	CDD18.3	3939	173	174	234	307	393	409	468	472	416	381	288	224
	CDH23.3	51694	1227	1389	2263	3770	5571	6152	7453	7453	6176	5094	3233	1912
	CDH26.7	29491	329	444	904	1934	3328	3864	4999	4982	3814	2817	1446	629
Wind	WSAvg	3.6	3.7	3.8	4.0	3.8	3.8	4.0	3.7	3.7	3.5	3.0	3.2	3.5
Precipitation	PrecAvg	61	13	4	2	4	1	0	1	1	0	2	23	10
	PrecMax	289	72	41	22	49	6	0	7	17	4	28	239	64
	PrecMin	4	0	0	0	1	1	0	0	0	0	0	0	0
	PrecStd	57	20	9	5	9	1	0	2	3	1	5	47	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	34.0	36.8	39.1	42.1	43.8	42.1	41.2	40.2	40.7	36.2	33.2
		MCWB	20.6	20.2	21.6	22.5	23.7	23.9	23.4	24.4	25.2	22.4	21.7	21.0
	2%	DB	31.0	32.1	34.2	37.0	39.2	40.1	40.8	39.9	38.2	37.8	34.8	32.1
		MCWB	21.4	21.0	21.7	23.0	23.9	23.8	23.7	25.4	26.0	24.5	23.0	21.5
	5%	DB	29.9	30.9	32.6	35.2	37.8	38.6	39.8	38.8	37.1	36.1	33.8	31.1
		MCWB	20.9	21.4	21.7	23.3	24.2	24.3	24.0	25.7	26.6	25.3	23.1	21.6
	10%	DB	28.7	29.5	31.0	34.0	36.1	37.1	38.7	37.8	36.1	35.0	32.5	30.0
		MCWB	20.6	21.0	21.6	23.1	24.1	24.8	24.4	26.3	26.9	25.4	23.2	21.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.9	25.2	25.8	27.2	29.1	29.0	29.8	31.1	30.6	29.1	27.0	25.5
		MCDB	29.2	30.1	31.5	33.7	36.5	36.2	35.8	36.0	35.8	34.1	32.2	29.9
	2%	WB	23.7	24.1	24.4	26.1	27.4	28.0	28.8	30.1	29.3	28.1	26.1	24.5
		MCDB	28.2	29.0	30.4	32.3	34.3	34.6	34.3	34.9	34.5	33.1	31.1	29.2
	5%	WB	22.9	23.2	23.6	25.2	26.5	27.3	28.0	29.3	28.9	27.4	25.2	23.8
		MCDB	27.3	27.9	29.6	31.6	33.5	33.7	33.9	34.2	34.0	32.5	30.3	28.6
	10%	WB	22.1	22.5	22.9	24.4	25.9	26.8	27.4	28.7	28.2	26.9	24.7	22.9
		MCDB	26.8	27.2	28.7	31.1	32.8	33.3	33.6	34.0	33.2	32.0	29.9	27.7
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.8	10.6	10.9	11.4	11.6	11.0	9.0	9.4	10.7	9.7	9.2
		MCDBR	10.8	11.3	12.5	12.4	13.4	13.7	12.9	10.8	10.7	12.2	11.0	10.5
		MCWBR	5.6	5.8	5.6	5.6	5.9	5.9	6.0	5.4	4.9	5.2	4.9	5.4
	5% WB	MCDBR	9.2	9.7	10.4	10.8	11.7	11.0	10.4	8.5	9.2	10.9	9.9	9.3
		MCWBR	5.3	5.4	5.6	5.8	6.0	5.6	5.8	5.0	4.6	5.0	4.6	5.0
Clear-Sky Solar Irradiance	taub	0.471	0.517	0.533	0.555	0.596	0.616	0.649	0.670	0.630	0.569	0.504	0.481	
	taud	2.085	1.951	1.902	1.877	1.772	1.727	1.700	1.666	1.723	1.852	2.014	2.060	
	Ebn at Noon	788	775	784	772	736	714	692	678	697	720	749	759	
	Edh at Noon	150	181	198	206	227	235	242	251	233	197	159	149	
All-Sky Solar Radiation	RadAvg	4.49	5.39	6.48	7.12	7.49	7.70	7.34	6.91	6.47	5.76	4.79	4.28	
	RadStd	0.24	0.24	0.21	0.25	0.22	0.15	0.19	0.17	0.19	0.16	0.20	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.59	+1.13	-1.80	N/A	-0.36	-0.27	N/A	N/A	+201	+201
(47)	Regional (2 neighbors)	+0.48	+0.70	-1.24	N/A	N/A	N/A	N/A	N/A	+188	+172

Nomenclature: See separate page